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Corporate Investment Under North Korea Threats*

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August 24, 2026

Abstract

We study how geopolitical risk from North Korea affects South Korean firms' investment. We construct a novel monthly GPRNK index from news articles, which spikes around nuclear tests and military confrontations and falls around summits. Heightened GPRNK reduces corporate investment, with larger declines among capital-intensive firms and smaller declines among firms with greater labor flexibility. Local projection estimates show that the decline in investment persists over time. The findings are consistent with both real options and Oi–Hartman–Abel channels in the transmission of geopolitical risk.

JEL Classification: D81, E22

Keywords: Geopolitical risk, Uncertainty, Textual analysis, Investment, Inter-Korean relations

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1 Introduction

A growing literature shows that heightened geopolitical risk delays firms' investment and may adversely affect macroeconomic outcomes (e.g., Caldara and Iacoviello, 2022; Ha et al., 2022; Wang et al., 2024). Building on this literature, we examine firm-level investment responses to fluctuations in inter-Korean relations. South Korea provides a useful setting for two reasons. First, tensions with North Korea have fluctuated repeatedly and often abruptly, ranging from military threats to sudden episodes of reconciliation, generating sharp variations in uncertainty that are plausibly exogenous to firms' decisions. Second, its economy relies heavily on export-oriented and capital-intensive industries, in which investment is particularly sensitive to uncertainty and costly to reverse.

We first develop the GPRNK index, a novel measure of geopolitical risk associated with inter-Korean relations, using automated keyword searches of articles from leading Korean-language news outlets. The index is constructed based on four main drivers of fluctuations in inter-Korean relations: military conflict, sanctions, talks, and economic engagement.

We next estimate the average effect of geopolitical risk on firm-level investment and find that heightened geopolitical risk reduces investment. A doubling of GPRNK, a change well within its observed range, reduces quarterly investment by 0.19 percentage points of assets, equivalent to 14% of the average investment rate. The effects remain significant after controlling for economic policy uncertainty and other macroeconomic determinants of investment, including domestic growth, the leading indicator, and world trade growth. We then examine heterogeneity by interacting GPRNK with firm characteristics and find significant differences in investment responses. Finally, using local projections (Jordà, 2005), we show that the decline in investment persists for several quarters after the initial shock.

We make three contributions to the literature. First, building on text-based measures of geopolitical risk (Caldara and Iacoviello, 2022), we construct a monthly geopolitical risk index focused on North Korea. This approach departs from earlier event studies based on discrete geopolitical incidents (Gerlach and Yook, 2016). Second, consistent with real options theory (Dixit and Pindyck,

1994), we find that North Korea-related geopolitical risk significantly reduces investment by South Korean firms. This result contributes to the broader literature on the transmission of political uncertainty to firm decisions (Gulen and Ion, 2016; Hassan et al., 2019), and the effects of geopolitical risk arising from proximity to military conflict (Federle et al., 2026). Third, we document heterogeneity in the transmission of geopolitical risk to investment. Capital-intensive firms experience larger declines, consistent with the real options channel, whereas firms with greater labor flexibility are less affected, consistent with the convexity channel (Hartman, 1972; Abel, 1983). Our study complements recent evidence highlighting flexible inputs as an adjustment margin under uncertainty (Bloom et al., 2024).

2 Measuring Geopolitical Risk from Inter-Korean Relations

Geopolitical risk on the Korean Peninsula reflects recurring cycles of conflict and reconciliation between South and North Korea, influenced by major powers in the region. This motivates our approach, which extends existing studies in two respects. First, we account for both the escalation of tensions and reconciliation. This allows us to track broader shifts in perceived uncertainty rather than focusing solely on crises. Second, we emphasize the interplay between geopolitical and economic developments. Military provocations can trigger sanctions, while diplomatic progress can facilitate economic cooperation.

The news articles are retrieved from BigKinds¹, a platform established by the Korea Press Foundation. We use 18 major South Korean newspapers and broadcasters and search for North Korea-related articles using keywords grouped into negative categories (military tensions and sanctions) and positive categories (talks/agreements and economic cooperation). To avoid misclassification, we exclude articles containing category-specific negation terms. For each outlet and period, we calculate the difference between the shares of negative and positive articles among all North Korea-related articles. The resulting series is scaled by its within-outlet standard deviation over 1995-2016. We then average across outlets to obtain a single time series and normalize it to a mean

¹<https://www.bigkinds.or.kr>

of 100 over the same period (see Appendix A.1 for details). As a sensitivity check, we also construct GPRNK-raw without excluding negation terms, and find a correlation of 0.98 with GPRNK.

Figure 1 plots the monthly GPRNK index from 1995 to 2024. The index rises around nuclear tests, missile launches, and military confrontations, and falls around bilateral or multilateral meetings. Compared with the Korea GPR index of Caldara and Iacoviello (2022), which captures broader, globally salient geopolitical risks, GPRNK focuses on North Korea-specific tensions. The two indices are only moderately correlated (0.33).

3 Empirical Framework and Data

To assess the effect of geopolitical risk on firm investment, we estimate:

$$Inv_{i,t+1} = \alpha_i + \delta_q + \theta GPRNK_t + X_t' \beta + C_{i,t}' \gamma + \epsilon_{i,t+1} \quad (1)$$

where $Inv_{i,t+1}$ is firm i 's investment expenditure in quarter $t + 1$ relative to total assets at the end of quarter t . X_t is a vector of aggregate controls for time-varying macroeconomic conditions, including Korea's economic policy uncertainty index, domestic GDP growth, the leading indicator, and world trade growth. $C_{i,t}$ is a vector of firm-level controls, including Tobin's q , leverage ratio, return on assets, asset size, and sales growth. We include firm fixed effects (α_i) and calendar quarter dummies (δ_q). Standard errors are two-way clustered by firm and year-quarter.

To examine heterogeneity, we extend Eq. (1) by interacting $GPRNK_t$ with proxies for capital intensity and factor substitutability: the fixed asset ratio, the labor-capital ratio, and the industry-level standard deviation of the labor-capital ratio.² The fixed asset ratio captures investment irreversibility associated with the real options channel, whereas the latter two variables reflect input flexibility underlying the convexity channel.

Next, we estimate local projections (Jordà, 2005) for horizons up to 12 quarters, using the same

²The industry-level measure captures cross-industry differences in input flexibility but may not fully reflect firm-specific substitution possibilities, consistent with the aggregation bias noted by Leahy and Whited (1996).

controls as in Eq. (1).

$$Inv_{i,t+h} = \alpha_i + \delta_q + \theta_h GPRNK_t + X_t' \beta_h + C_{i,t}' \gamma_h + \varepsilon_{i,t+h}, \quad h = 0, 1, \dots, 12, \quad (2)$$

where $Inv_{i,t+h}$ is the investment rate of firm i at horizon h . θ_h traces the response of investment at horizon h to a one-unit increase in GPRNK. For heterogeneity, we add the same interactions as above. The specification otherwise follows Eq. (1).

While geopolitical shocks originating from North Korea are plausibly exogenous, we further address concerns that GPRNK may capture endogenous media attention or broader macroeconomic conditions using two-stage least squares (2SLS). Following Ha et al. (2022), we instrument GPRNK with negative North Korea-related events (Appendix A.2). We also conduct robustness checks using subtopic indices and subsample analyses.

Our empirical analysis uses a sample of 2,055 South Korean non-financial firms from 2000 to 2024. Firm-level accounting data are obtained from KIS Value, provided by National Information & Credit Evaluation (NICE). Table 2 reports summary statistics, and Appendix A.3 describes the variables.

4 Results

On average, geopolitical risk has a significant adverse effect on South Korean firms' investment (Table 3). The baseline estimate is -0.0028. Since we use the log of GPRNK, a doubling of the index implies a decline of about 0.19 percentage points in investment as a share of assets, equivalent to 14% of the average investment rate. This magnitude is comparable to the 14% decline reported by Wang et al. (2024) for U.S. firms. The estimates remain robust to the inclusion of aggregate controls and 2SLS estimation.

Table 4 shows heterogeneous investment responses. Firms with higher fixed-asset ratios have larger declines in investment as GPRNK rises. The negative effect of GPRNK on investment is attenuated for firms with higher labor-to-capital ratios. The estimated interaction term for labor-

capital substitutability has the expected sign but is not statistically significant.

Figure 2 shows that an increase in GPRNK is followed by an immediate decline in investment that persists for up to three years. The heterogeneous responses are consistent with the main regression results. The IV estimates in Figure A.4 show a similar dynamic pattern.

Because GPRNK captures both escalations and easing of geopolitical tensions, Table 5 decomposes the index into negative and positive components and their subtopics. The negative component significantly reduces investment, with both threats and sanctions having sizable adverse effects. By contrast, the positive component is insignificant on average: diplomatic talks have no discernible effect, whereas inter-Korean economic cooperation significantly increases investment. These findings are consistent with Segal et al. (2015) on the asymmetric effects of positive and negative uncertainty innovations.

Table 6 reports robustness checks across sectors and over time. The baseline results hold for both manufacturing and non-manufacturing firms. For the time dimension, we use a smooth transition model to identify a transition ending in 2014Q3 (Appendix A.4) and compare the investment effects before and after the transition. The effect is negative and significant before the transition but close to zero thereafter, with a pooled interaction test confirming a significantly weaker post-transition effect. Splitting the sample by ruling party, we find that the estimated coefficients are negative under both conservative and progressive governments, but only the former is statistically significant.

5 Conclusions

We develop a new index of geopolitical risk specific to inter-Korean relations, capturing both escalation and reconciliation episodes. Geopolitical risk significantly reduces corporate investment, with stronger effects for capital-intensive firms and weaker effects for firms with higher labor–capital ratios, consistent with real options and the Oi–Hartman–Abel channels, respectively. Local projection estimates show that the decline in investment persists for several quarters.

Amid heightened geopolitical fragmentation, our findings have two broader implications. The

greater sensitivity of capital-intensive firms to geopolitical risk underscores the importance of a stable and predictable policy environment for long-term investment, particularly in economies reliant on such industries. The weaker effects among firms with greater input flexibility suggest that policies that reduce frictions in input adjustment may strengthen firms' resilience to geopolitical shocks.

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Use of AI tools

OpenAI ChatGPT (GPT-5.6 Sol) was used to correct grammatical and typographical errors. The authors reviewed the final text and take full responsibility for the content.

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Figures

FIGURE 1 The GPRNK Index (1995-2024)

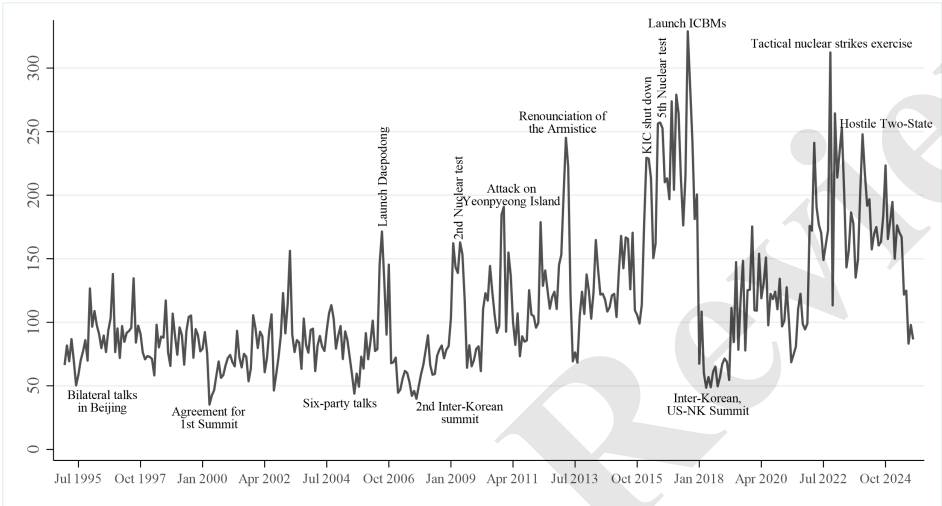
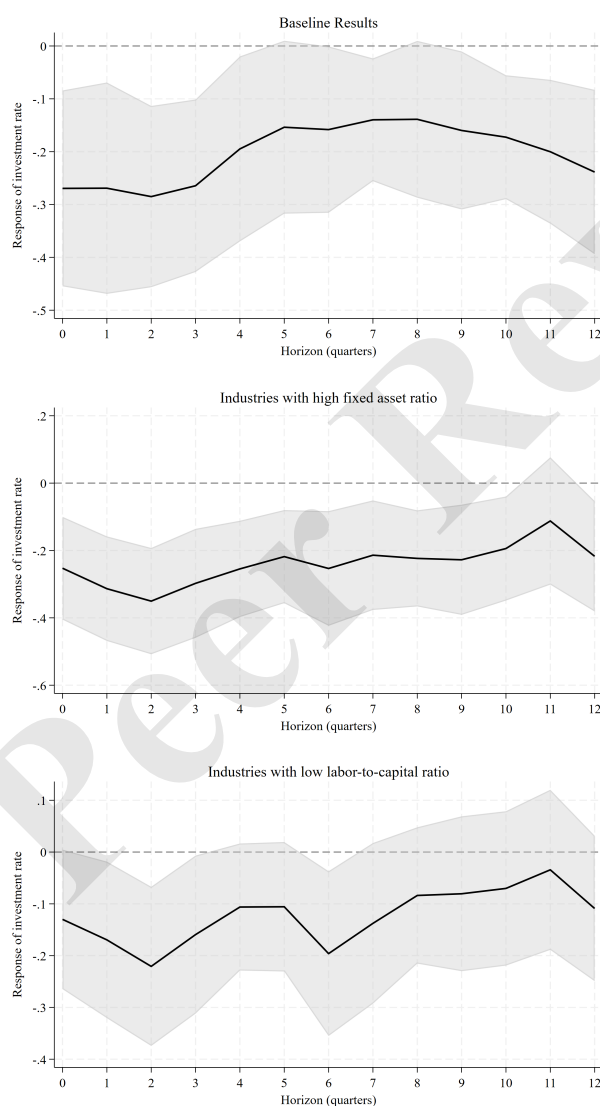


FIGURE 2 Impulse response functions from local projection



Notes: IRFs to a one-unit increase in log GPRNK over 12 quarters. Y-axis in percentage points. Shaded bands denote 95% CIs.

Tables

TABLE 1 Search keywords

Category		Topic	Action/Status	Excluded	Number of Articles
Default keywords		North Korea			1,441,420
Negative	Military tensions	(nuclear) or (missile) or (military) or (war)	(threat) or (tension) or (provocation)	(peace)	229,361
	Sanctions	(sanction) or (pressure)	(refute) or (dissent) or (criticize)	-	45,642
Positive	Talks/agreements	(talks) or (dialogue)	(resume) or (agreement) or (negotiation)	(fail) or (break) or (boycott)	250,639
	Economic cooperation	(economic cooperation) or (abbreviation of economic cooperation)	(progress) or (expectation)	(concern)	40,601

Notes: See Appendix Table A1 for Original Korean search keywords.

TABLE 2 Descriptive statistics

	Mean	SD	Min	Max	Obs
Uncertainty measures and macro variables					
GPRNK (log)	4.79	0.42	3.84	5.62	122,622
EPU (log)	5.04	0.41	4.02	5.93	122,622
GDP growth rate (QoQ, %)	0.76	0.89	-3.4	3.1	122,622
Composite index of leading indicator (log)	4.61	0.01	4.57	4.63	122,622
World merchandise trade growth rate (QoQ, %)	0.64	3.05	-16.08	13.31	122,622
Firm characteristics					
Investment expenditures	0.01	0.05	-9.86	9.47	122,622
Tobin's q	1.47	1.57	0.20	108.55	120,083
Leverage ratio	0.24	0.17	0.00	1.57	120,083
ROA	0.00	0.06	-0.94	10.46	120,808
Asset size (log)	25.76	1.31	20.01	33.41	121,430
Sales growth	0.01	0.42	-14.07	13.82	118,233
Capital intensity	0.52	0.19	0.00	1.00	121,294
Labor to capital ratio (log)	-18.58	1.33	-24.15	-4.54	119,217
Labor-capital substitutability	1.04	0.37	0.00	4.20	120,984

TABLE 3 GPRNK and firms' investment: average effects

	(1) OLS	(2) OLS	(3) OLS	(4) IV
GPRNK	-0.0028*** (0.0010)	-0.0020* (0.0011)	-0.0019* (0.0010)	-0.0030* (0.0016)
EPU		-0.0025** (0.0010)	-0.0027** (0.0011)	-0.0022* (0.0012)
GDP growth			0.0006 (0.0004)	0.0005 (0.0004)
Leading indicator			-0.0581 (0.0372)	-0.0550 (0.0381)
World trade growth rate			0.0064 (0.0126)	0.0066 (0.0131)
Tobin's q	0.0007*** (0.0002)	0.0006** (0.0002)	0.0007** (0.0003)	0.0007** (0.0003)
Leverage ratio	-0.0048** (0.0020)	-0.0051*** (0.0019)	-0.0056*** (0.0019)	-0.0058*** (0.0019)
ROA	0.0112*** (0.0041)	0.0105** (0.0041)	0.0106** (0.0041)	0.0103** (0.0041)
Asset size	-0.0090*** (0.0007)	-0.0085*** (0.0007)	-0.0084*** (0.0007)	-0.0082*** (0.0007)
Sales growth	0.0003 (0.0003)	0.0003 (0.0003)	0.0003 (0.0003)	0.0003 (0.0003)
Firm fixed effects	Yes	Yes	Yes	Yes
Calendar quarter dummies	Yes	Yes	Yes	Yes
Number of obs.	117,934	117,934	117,934	117,934
Number of firms	2,047	2,047	2,047	2,047
Adj. R^2	0.0741	0.0743	0.0745	0.0116
K-P Underidentification test p-value				0.0005
K-P Weak identification test Wald F				14.23

Notes: We regress firm-level investment rate on the GPRNK index. The dependent variable is investment expenditures in quarter $t + 1$ divided by asset size at the end of quarter t . The GPRNK is the logarithm of the 3-month average GPRNK index in quarter t . Standard errors are two-way clustered at the firm and year-quarter levels, and reported in parentheses. *, **, *** indicates statistical significance at the 10%, 5%, 1% level, respectively. We control Tobin's q, leverage ratio, return on asset ratio, natural logarithm of asset size, and sales growth rate. In column (4), we present two-stage least squared estimates using the number of North Korea-related events as an instrument variable. The sample covers 2003q2-2024q4. K-P denotes Kleibergen-Paap; the underidentification test is based on the rk LM statistic.

TABLE 4 GPRNK and firms' investment: differential effects

	(1) OLS	(2) OLS	(3) IV	(4) OLS	(5) OLS	(6) IV	(7) OLS	(8) OLS	(9) IV
GPRNK	0.0012 (0.0009)			0.0150** (0.0054)			-0.0039*** (0.0016)		
GPRNK * Capital intensity	-0.0011** (0.0004)	-0.0012*** (0.0004)	-0.0014*** (0.0006)						
Capital intensity	0.0050** (0.0023)	0.0059** (0.0023)	0.0065** (0.0032)						
GPRNK * Labor-to-capital				0.0012*** (0.0004)	0.0013*** (0.0004)	0.0020*** (0.0006)			
Labor-to-capital				-0.0064*** (0.0022)	-0.0071*** (0.0022)	-0.00102*** (0.0027)			
GPRNK * L-K substitutability							0.0018 (0.0013)	0.0010 (0.0011)	0.0024 (0.0016)
L-K substitutability							-0.0108* (0.0063)	-0.0056 (0.0054)	-0.0127* (0.0076)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Number of observations	117,934	117,934	117,934	116,086	116,086	116,086	117,516	117,516	117,516
Number of firms	2,047	2,047	2,047	2,047	2,047	2,047	2,044	2,044	2,044
Adj. R^2	0.0748	0.0800	0.0029	0.0738	0.0790	0.0029	0.0745	0.0796	0.0027
K-P χ^2 p-value			0.0001			0.0001			0.0001
K-P Wald F			20.26			20.50			24.11

Notes: We regress firm-level investment rate on geopolitical risk from North Korea conditioned on capital intensity and convexity of the marginal product of capital. Standard errors are two-way clustered at the firm and year-quarter levels, and reported in parentheses. *, **, *** indicates statistical significance at the 10%, 5%, 1% level, respectively. We control Tobin's q, leverage ratio, return on asset ratio, natural logarithm of asset size, and sales growth rate. We include the first order term of GPRNK index in column (1), (4), (7), while we control time fixed effects thereby dropping the GPRNK term in column (2), (5), (8). In column (3), (6), (9), we present two-stage least squared estimates with two-way fixed effects specification. The sample covers 2003q2 2024q4. K-P denotes Kleibergen-Paap; the χ^2 p-value is from the underidentification test, and the Wald F-statistic is for weak identification test.

TABLE 5 GPRNK and firms' investment: subtopic indices

	(1) Negative	(2) Positive	(3) Threat	(4) Sanction	(5) Talks	(6) Economic cooperation
Panel A: average effects						
Subtopic index	-0.0030*** (0.0007)	0.0017 (0.0011)	-0.0026** (0.0007)	-0.0028*** (0.0006)	0.0013 (0.0011)	0.0024*** (0.0007)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Calendar quarter dummies	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	117,928	117,928	117,928	117,928	117,928	117,928
Adjusted R^2	0.0736	0.0743	0.0742	0.0744	0.0736	0.0742
Panel B: differential effects						
Subtopic index * capital intensity	-0.0007** (0.0003)	0.0007* (0.0004)	-0.0007** (0.0003)	-0.0008*** (0.0003)	0.0006 (0.0004)	0.0009*** (0.0003)
Capital intensity	0.0035* (0.0016)	-0.0035** (0.0018)	0.0032** (0.0015)	0.0041*** (0.0014)	-0.0028 (0.0018)	-0.0035*** (0.0011)
Subtopic index * Labor-to-capital	0.0010*** (0.0003)	-0.0007 (0.0005)	0.0009*** (0.0003)	0.0009*** (0.0003)	-0.0006 (0.0004)	-0.0007** (0.0003)
Labor-to-capital	-0.0058*** (0.0015)	0.0022 (0.0020)	-0.0050*** (0.0014)	-0.0053*** (0.0015)	0.0018 (0.0020)	0.0019* (0.0010)
Subtopic index * Labor-capital substitutability	-0.0002 (0.0009)	-0.0006 (0.0012)	-0.0001 (0.0008)	-0.0010 (0.0009)	-0.0005 (0.0011)	-0.0003 (0.0008)
Labor-capital substitutability	0.0016 (0.0051)	-0.0000 (0.0046)	-0.0004 (0.0042)	0.0039 (0.0044)	0.0012 (0.0050)	0.0002 (0.0031)
Firm and time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	115,699	115,699	115,699	115,699	115,699	115,699
Number of firms	2,044	2,044	2,044	2,044	2,044	2,044
Adjusted R^2	0.0785	0.0787	0.0786	0.0787	0.0785	0.0787

Notes: We regress firm-level investment rate on subtopic indices. We estimate average effects in the top panel and differential effects conditioned on capital intensity and convexity of the marginal product of capital in the bottom panel. Standard errors are two-way clustered at the firm and year-quarter levels, and reported in parentheses. *, **, *** indicates statistical significance at the 10%, 5%, 1% level, respectively. We control Tobin's q, leverage ratio, return on asset ratio, natural logarithm of asset size, and sales growth rate. We include calendar quarter dummies in the top panel, while we control time fixed effects thereby dropping the GPRNK term in the bottom panel. The sample covers 2003q2-2024q4.

TABLE 6 GPRNK and firms' investment: subsample analysis

	(1) Manufacturing	(2) Non manufacturing	(3) Through 2014Q3	(4) From 2014Q4	(5) Progressive ruling party	(6) Conservative ruling party
GPRNK	-0.0030** (0.0011)	-0.0021** (0.0009)	-0.0054*** (0.0017)	0.0003 (0.0002)	-0.0016 (0.0013)	-0.0071*** (0.0015)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Calendar quarter dummies	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Number of obs.	81,417	36,517	49,880	68,033	51,348	66,570
Adjusted R^2	0.0557	0.2266	0.0325	0.3226	0.0190	0.2552

Notes: Standard errors are two-way clustered at the firm and year-quarter levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. We include price-to-book value ratio (Tobin's q), leverage ratio, ROA and asset size as firm-level control variables. The sample covers 2003q2-2024q4. Columns (3)–(4) split the sample at 2014Q3, the estimated end of the transition period from a smooth-transition AR(2) model of GPRNK index.

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Appendix

August 24, 2026

Abstract

We study how geopolitical risk from North Korea affects South Korean firms' investment. We construct a novel monthly GPRNK index from news articles, which spikes around nuclear tests and military confrontations and falls around summits. Heightened GPRNK reduces corporate investment, with larger effects for capital-intensive firms and weaker effects for firms with greater labor flexibility. Local projection estimates show that the decline in investment persists over time. The findings are consistent with both real options and Oi–Hartman–Abel channels in the transmission of geopolitical risk.

JEL Classification: D81, E22

Keywords: Geopolitical risk, Uncertainty, Textual analysis, Investment, Inter-Korean relations

Appendix

A.1.The GPRNK index

We go through five steps in selecting the search keywords. First, we set “North Korea” as a default keyword to pick up any article associated with the geopolitical risk of inter-Korean relations.¹ Second, within the set of articles selected in the first step, we flag articles that mention one of four types of geopolitical risk, reflecting the main drivers of inter-Korean relations and their potential economic impacts on South Korea. Those are military tensions, sanctions, talks/agreements, and economic cooperation. Third, we start with a human reading of articles around the time of the major geopolitical events and list keywords regarding the subjects of those events (the topic) and the descriptions of the subjects (the action/status). Fourth, we list the words that collocate with the topic but negate the original aspects of the events, and we exclude those from the search keywords. By doing so, we avoid falsely finding articles that report the opposite. Lastly, we finalize the words through iterations of the validation process to select the ones that recur and sufficiently cover the geopolitical events in the categories over time. Table A.1 shows the search keywords in Korean.

TABLE A.1 Search keywords

Category		Topic	Action/Status	Excluded	Number of Articles
Default keywords		North Korea			1,441,420
Negative	Military tensions	(핵) or (미사일) or (군사) or (전쟁)	(위협) or (긴장) or (도발)	(평화)	229,361
	Sanctions	(제재) or (압박)	(반발) or (불복) or (비난)	-	45,642
Positive	Talks/agreements	(대화) or (회담)	(재개) or (합의) or (협약)	(결렬) or (무산) or (거부)	250,639
	Economic cooperation	(경제협력) or (경협)	(추진) or (기대)	(우려)	40,601

Next, we compute the frequency of news articles so that $N_{j,it}$ denotes the total number of articles containing the keywords in each category j from media i at time t . The four categories—military threats, sanctions, talks, and economic cooperation—can be combined into two broader categories: negative ($N_{neg,it}$) and positive ($N_{pos,it}$). The relative frequency of net negative news articles compared to the total number of news articles related to North Korea is computed as follows:

$$X_{it} = \frac{N_{neg,it} - N_{pos,it}}{N_{it}} \quad (1)$$

where X_{it} denotes the relative frequency of net negative news articles for media i and time t . N_{it}

¹Note that South Korean newspapers often use Chinese characters (北, or 北韓) when referring to North Korea.

is the total number of articles with the default search keyword “North Korea”. After transforming X_{it} into a positive-valued series, $\widetilde{X}_{it}$,² we standardize it to obtain a series Y_{it} , with a unit standard deviation for each media outlet.

$$Y_{it} = \frac{\widetilde{X}_{it}}{\sigma_i} \quad (2)$$

where σ_i is the within-outlet standard deviation of $(\widetilde{X}_{it})$ from January 1995 to December 2016. The standardized index is averaged across outlets.

$$Y_t = \frac{1}{N} \sum_{i=1}^N Y_{it} \quad (3)$$

where N is the total number of media outlets. Then we normalize Y_t , such that $GPRNK_t = \frac{100Y_t}{\bar{Y}}$ where $\bar{Y}$ is the mean value of Y_t from 1995 to 2016.

In Figure A.1, we check the robustness of the GPRNK index by examining potential biases due to sample coverage.³ Media bias may undermine the representativeness of our index. If one finds systematic differences in the GPRNK index due to an outlet’s political slant, the index produced by a simple average of individual media could fail to represent unbiased media attention on North Korea-related issues in South Korea.

We compute the subgroup GPRNK indices by the political leaning of the news outlet (top panel).⁴ The indices computed from right-leaning newspapers and the one from left-leaning newspapers show close co-movement, with a correlation coefficient of 0.94. In addition, we compare the GPRNK index with its sub indices constructed by press type: daily newspapers, economic newspapers, and broadcast media (bottom panel). The correlations are uniformly high, ranging from 0.94 to 0.97.

Next, we compare our index with two other existing indices that quantify geopolitical risks from North Korea (Figure A.2). First, we compare the GPRNK index with the GPR Korea index by Caldara and Iacoviello (2022). The GPR Korea index uses automated text searches to count the number of articles related to geopolitical risk in Korea. The search query of the GPR Korea index is a mixture of general keywords applicable to worldwide geopolitical risk.⁵ Therefore, it is suitable for capturing broader geopolitical risks of global interest, such as the threat of war, terrorism, or

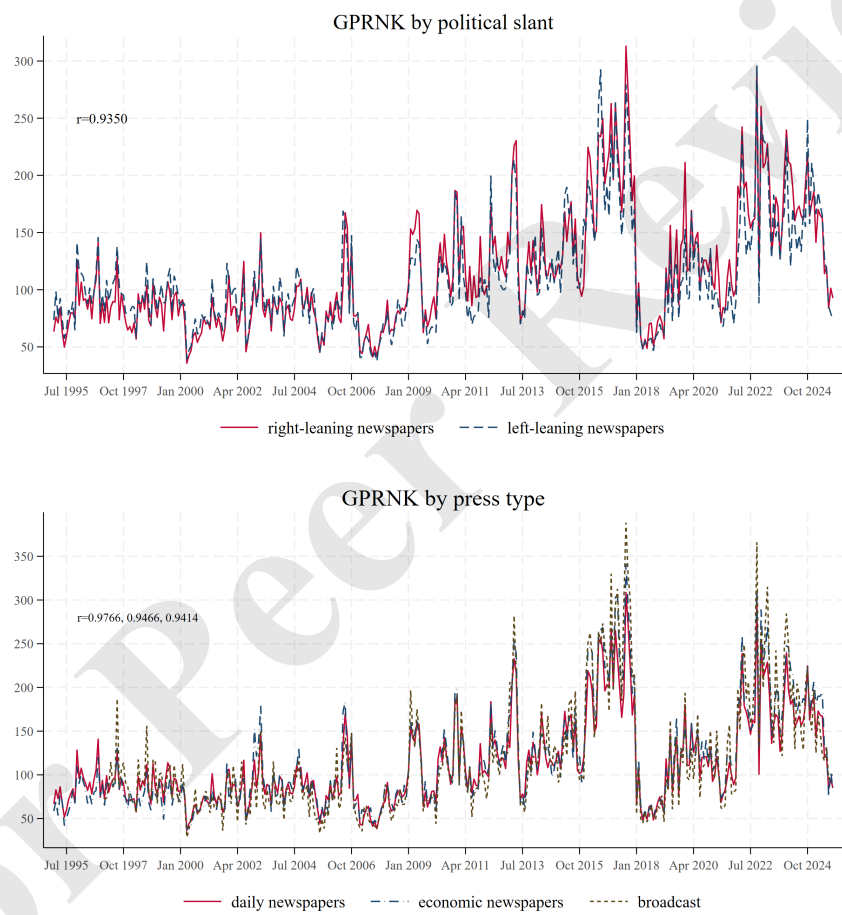
$$^2 \widetilde{X}_{it} = \frac{1}{2} \left\{ X_{it} + \sqrt{X_{it}^2 + 0.1} \right\}$$

³Our media sample covers 68.9% of total newspaper circulation and 63.6% of total viewership among national broadcast channels.

⁴Right-leaning newspapers are the Chosun Ilbo, Dong-a Ilbo, Joong-ang Ilbo, Kookmin Ilbo and Munhwa Ilbo. Left-leaning newspapers are the Kyunghyang Sinmun and Hankyoreh.

⁵The search query for the GPR Korea index is “Korea AND (tensions/risk/fear/chaos/uncertainty/unrest/violence...) AND (military/war/geopolitical/coup/guerrilla/warfare/army/terrorism)”. For details of the methodology and trends of the GPR Korea index, refer to their website (<https://www.matteoiacoviello.com/gpr.htm>).

FIGURE A.1 Assessing Potential Selection Bias of the GPRNK Index



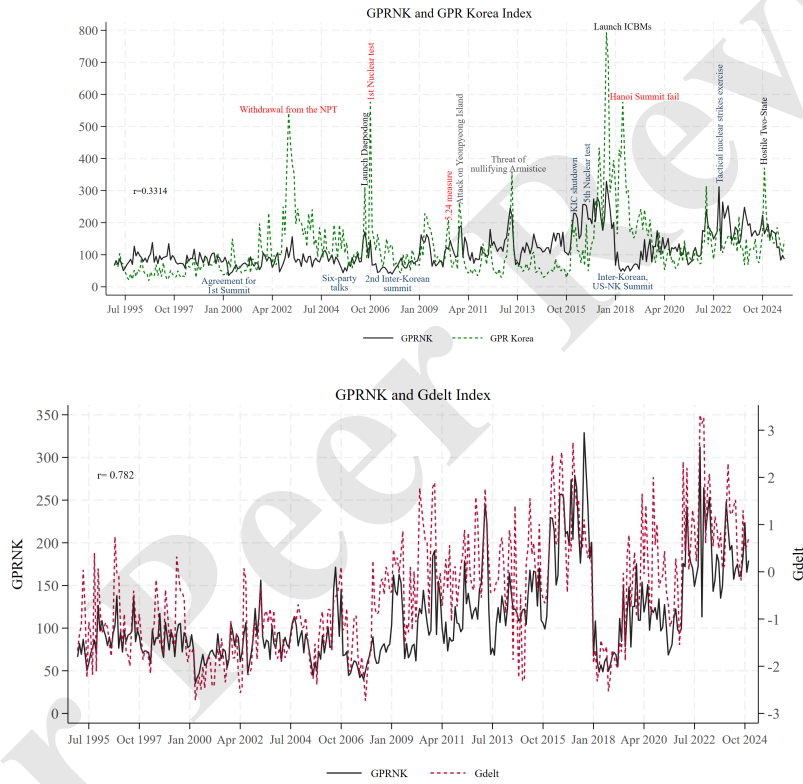
cross-border tensions, but it may fail to capture the Korea-specific context, such as nuclear weapons development or bilateral talks. Furthermore, it may reflect a perspective centered on the U.S., the U.K., and Canada, as its press coverage is from 11 leading English-language newspapers published in those countries. The GPR Korea index peaks around the time of North Korea's withdrawal from the Non-Proliferation Treaty and around the time when North Korea threatened to nullify the 1953 armistice accord, both of which were more significant for the landscape of global diplomacy than for the inter-Korea relationship. However, compared with our index, the GPR Korea index fails to capture important fluctuations in geopolitical risk, including the reduction in geopolitical risk during the summit meetings and its stepwise escalation in 2016 and 2017. The correlation between the GPR Korea index and the GPRNK index is approximately 0.33.

Second, we compare our index with the GDELT-based measure of inter-Korean relations (in the bottom panel of Figure A.1). The GDELT-based measure is constructed using the Global Database of Events, Language, and Tone (GDELT)⁶, which records dyadic events between countries and classifies each event by the acting country, the target country, and the nature of the interaction. Focusing on events between South Korea and North Korea, we utilize the Goldstein scale, which assigns each event a score ranging from -10 (most conflictual) to +10 (most cooperative). Specifically, we compute the monthly average Goldstein score for events in which South Korea is the acting country and North Korea is the target, as well as for events in which North Korea is the acting country and South Korea is the target. The overall GDELT-based inter-Korean relations index is then defined as the simple average of these two directional measures, capturing the prevailing tone of bilateral interactions between the two Koreas. Since the GDELT index increases when bilateral relations improve, we multiply it by -1 so that higher values consistently indicate heightened geopolitical risk.

The GDELT-based index primarily focuses on bilateral relationships, limiting its ability to fully capture the complex multi-actor regional risks posed by North Korea. By contrast, the GPRNK index is, by design, constructed to reflect region-wide geopolitical risk that extends beyond a single bilateral relationship, drawing on targeted textual analysis. Moreover, it explicitly distinguishes between positive events—such as diplomatic dialogue or economic cooperation—and negative events—such as military threats or sanctions—thereby enhancing transparency and interpretability. Despite these conceptual differences, the GPRNK and GDELT-based inter-Korean relations indices are strongly correlated, with a correlation coefficient of 0.78, indicating that both measures capture common underlying geopolitical dynamics.

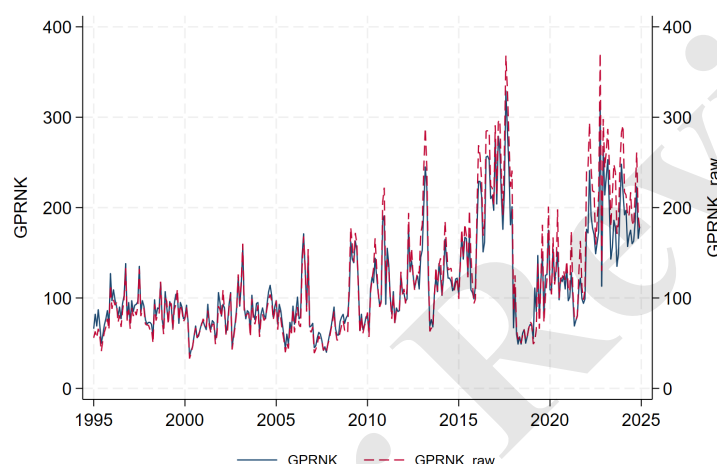
⁶See <https://www.gdeltproject.org/> for the details.

FIGURE A.2 Comparing the GPRNK Index with Existing Measures



Finally, as a check on the sensitivity of the index to keyword selection, we construct an alternative series, GPRNK-raw, using the same searches without excluding articles that contain negation terms. Figure A.3 compares GPRNK with GPRNK-raw and shows that the two series are highly correlated (0.98).

FIGURE A.3 GPRNK Index under Alternative Keyword Searches



A.2. Construction of the Event-Based Instrument

We construct the instrument from a hand-collected chronology of North Korea-related events spanning 2003-2024, drawn from the Republic of Korea Ministry of Unification and the Arms Control Association, and cross-validated against the event lists in Kim and Roland (2014) and Ha et al. (2022). We restrict our attention to events related to North Korea's military threat. Ordinary rhetorical threats are excluded, and the event is retained only if it involves a concrete, verifiable action, such as a missile launch, a nuclear test, or an armed skirmish. We make two sets of exceptions to this rule. First, we retain declarations that mark a substantive turning point in North Korea's nuclear posture, even though they are not accompanied by contemporaneous physical actions; specifically, the declaration of withdrawal from the Non-Proliferation Treaty, the declaration of possessing nuclear weapons, and the declaration of lifting the moratorium on nuclear testing. Second, we retain the August 2017 exchange between Trump's "fire and fury" statement and North Korea's announcement of a plan to strike near Guam, given the exceptional international salience of this episode as a bilateral escalation event.

We aggregate the resulting event count to quarterly frequency and use it, $Event_t$, as the ex-

cluded instrument for $GPRNK_t$ in the two-stage least squares specification. Because these events reflect discrete, verifiable actions and policy declarations whose timing is determined by North Korea's strategic decisions rather than by the outcomes of individual South Korean firms, we argue that $Event_t$ satisfies the exclusion restriction. The full list of events included in the construction of the instrument is presented in Table A.2. Figure A.4 plots the local projection results.

TABLE A.2 North Korea-related Events Timeline (2003-2024)

Date	Event	Reference
2003-01-10	NK announces withdrawal from NPT	armscontrol association
2003-02-24	NK fires 1 missile into East Sea	Ministry of Unification
2003-02-24	NK test-fires anti-ship cruise missile	arms control
2003-03-10	NK fires surface-to-ship missile off east coast	Ministry of Unification
2004-11-01	3 NK patrol boats cross NLL near Socheong/Yeonpyeong Islands, retreat after warning shots	armscontrol association
2005-02-10	NK Foreign Ministry declares it has produced nuclear weapons	armscontrol association
2005-03-02	NK states it is no longer bound by missile test moratorium	armscontrol association
2006-07-05	NK test-fires 7 missiles including Taepodong-2	armscontrol association
2006-10-09	NK conducts first nuclear test	armscontrol association
2008-03-28	NK conducts missile tests	armscontrol association
2009-04-05	NK launches Unha-2 rocket	armscontrol association
2009-04-14	NK declares withdrawal from Six-Party Talks	armscontrol association
2009-05-25	NK conducts second nuclear test	armscontrol association
2010-03-26	ROKS Cheonan sinking	armscontrol association
2010-11-23	NK shells Yeonpyeong Island	armscontrol association
2011-02-28	NK threatens to turn Seoul into 'sea of fire' in response to joint exercises	armscontrol association
2012-04-13	NK's Unha-3 satellite/rocket launch fails	armscontrol association
2012-12-12	NK successfully launches Unha-3	armscontrol association
2013-02-12	NK conducts third nuclear test	armscontrol association
2013-04-09	NK threatens to shut Kaesong complex, warns South to prepare for foreigner evacuation	Ministry of Unification
2013-05-18	NK fires 5 short-range projectiles off east coast	Ministry of Unification
2014-02-21	NK fires 4 short-range projectiles	Ministry of Unification
2014-02-27	NK fires 4 ballistic missiles	Ministry of Unification
2014-03-26	NK test-fires 2 medium-range Rodong missiles, violating UN sanctions	armscontrol association
2014-05-22	NK fires 2 rounds near ROK vessel south of NLL; ROK returns fire with 5 rounds	Ministry of Unification
2014-06-26	NK fires 3 short-range projectiles into East Sea	Ministry of Unification
2014-06-28	NK fires 2 short-range ballistic missiles	Ministry of Unification
2014-07-02	NK fires 2 short-range projectiles into East Sea	Ministry of Unification
2014-07-09	NK fires 2 short-range ballistic missiles into East Sea	Ministry of Unification
2014-07-13	NK fires 2 short-range missiles into East Sea	Ministry of Unification
2014-07-26	NK fires 1 short-range missile into East Sea	Ministry of Unification
2014-07-28	NK fires 4 short-range projectiles eastward from Mt. Myohyang	Ministry of Unification
2014-09-01	NK fires 1 short-range projectile from Jagang Province into East Sea	Ministry of Unification
2014-09-06	NK fires 3 short-range projectiles from Wonsan into East Sea	Ministry of Unification
2014-10-07	NK patrol boat crosses NLL near Yeonpyeong, sparking first inter-Korean naval clash in 5 years	Ministry of Unification
2015-02-07	NK test-fires new anti-ship rocket	Ministry of Unification
2015-02-08	NK fires 5 short-range missiles into East Sea	Ministry of Unification
2015-03-02	NK fires 2 short-range missiles into East Sea	Ministry of Unification
2015-03-12	NK conducts surface-to-air missile drill, fires 7 missiles from Sondok	Ministry of Unification

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TABLE A.2 North Korea-related Events Timeline (2003-2024) – continued from previous page

Date	Event	Reference
2015-06-14	NK fires 3 KN-01 short-range missiles into East Sea	Ministry of Unification
2015-08-20	NK fires artillery near ROK loudspeakers	Ministry of Unification
2015-11-28	NK's SLBM test fails	Ministry of Unification
2016-01-06	NK conducts fourth nuclear test, claims hydrogen bomb	armscontrol association
2016-02-07	NK launches long-range rocket described as satellite launch	armscontrol association
2016-03-03	NK fires 6 short-range missiles from Wonsan into East Sea	Ministry of Unification
2016-03-10	NK fires 2 short-range ballistic missiles from Sakkanmol into East Sea	Ministry of Unification
2016-03-18	NK fires 2 medium-range ballistic missiles from Sukchon into East Sea	Ministry of Unification
2016-04-01	NK fires 3 short-range surface-to-air missiles from Sondok into East Sea	Ministry of Unification
2016-04-15	NK test-fires 1 missile off east coast, presumed failure	Ministry of Unification
2016-04-15	NK test-launches intermediate-range Musudan missile (failure)	armscontrol association
2016-04-28	NK fires missiles from Wonsan twice, both presumed failures	Ministry of Unification
2016-05-28	NK fires 1 missile from Wonsan, presumed failure	Ministry of Unification
2016-06-22	NK fires 2 ballistic missiles from Wonsan	Ministry of Unification
2016-06-22	NK conducts two more Musudan tests (partial success)	armscontrol association
2016-07-09	NK fires 1 SLBM from waters off Sinpo	Ministry of Unification
2016-07-19	NK fires 3 short-range ballistic missiles from Hwangju into East Sea	Ministry of Unification
2016-08-03	NK fires 1 ballistic missile from Unryul into East Sea	Ministry of Unification
2016-08-24	NK test-fires 1 SLBM from waters near Sinpo into East Sea	Ministry of Unification
2016-09-05	NK fires 3 ballistic missiles from North Hwanghae into East Sea	Ministry of Unification
2016-09-09	NK conducts fifth nuclear test	armscontrol association
2017-02-12	NK fires 1 ballistic missile from Panghyon	Ministry of Unification
2017-02-12	NK tests Pukguksong-2 ballistic missile	armscontrol association
2017-05-14	NK tests the Hwasong-12 missile. The IRBM test is successful.	Ministry of Unification
2017-05-21	NK fires 1 Pukguksong-2 IRBM	Ministry of Unification
2017-05-28	NK fires presumed Scud-variant (KN-18) SRBM from Wonsan	Ministry of Unification
2017-07-04	NK tests Hwasong-14, its first ICBM-range missile	armscontrol association
2017-07-28	NK conducts second Hwasong-14 test with extended range	armscontrol association
2017-08-09	Trump 'fire and fury' remarks; NK announces plan to strike near Guam	armscontrol association
2017-08-26	NK fires several SRBMs into East Sea	Ministry of Unification
2017-08-29	NK fires Hwasong-12 missile that overflies Japan	armscontrol association
2017-09-03	NK conducts sixth (largest) nuclear test, claims hydrogen bomb	armscontrol association
2017-09-15	NK fires 1 Hwasong-12 missile from Sunan into East Sea	Ministry of Unification
2017-09-19	Trump threatens to 'totally destroy North Korea' at UN General Assembly	armscontrol association
2017-11-29	NK tests Hwasong-15 ICBM	armscontrol association
2019-05-04	NK tests new short-range missiles, first since 2017	armscontrol association
2019-05-09	NK fires two additional short-range ballistic missiles	armscontrol association
2019-07-25	NK resumes testing of KN-23 short-range ballistic missile	armscontrol association
2019-07-31	NK fires 2 SRBMs from Hodo Peninsula	Ministry of Unification
2019-08-06	NK fires 2 presumed SRBMs near Gwail County	Ministry of Unification
2019-08-10	NK fires 2 presumed SRBMs from Hamhung	Ministry of Unification
2019-08-24	NK fires 2 unidentified projectiles, presumed SRBMs, from Sondok	Ministry of Unification
2019-09-10	NK fires unidentified missiles twice eastward; Kim Jong Un oversees super-large MLRS test	Ministry of Unification
2019-10-02	NK tests new SLBM Pukguksong-3	armscontrol association
2019-11-28	NK fires 2 unidentified projectiles from Yonpo into East Sea	Ministry of Unification
2020-01-01	Kim announces end of test moratorium, pursuit of 'new strategic weapon'	armscontrol association
2020-03-02	NK conducts a series of short-range projectile tests in March	armscontrol association
2020-03-09	NK fires 3 unidentified projectiles from Sondok into East Sea	Ministry of Unification
2020-03-21	NK fires 2 projectiles from North Pyongan into East Sea	Ministry of Unification
2020-03-28	NK fires 2 projectiles from Wonsan into East Sea	Ministry of Unification

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TABLE A.2 North Korea-related Events Timeline (2003-2024) – continued from previous page

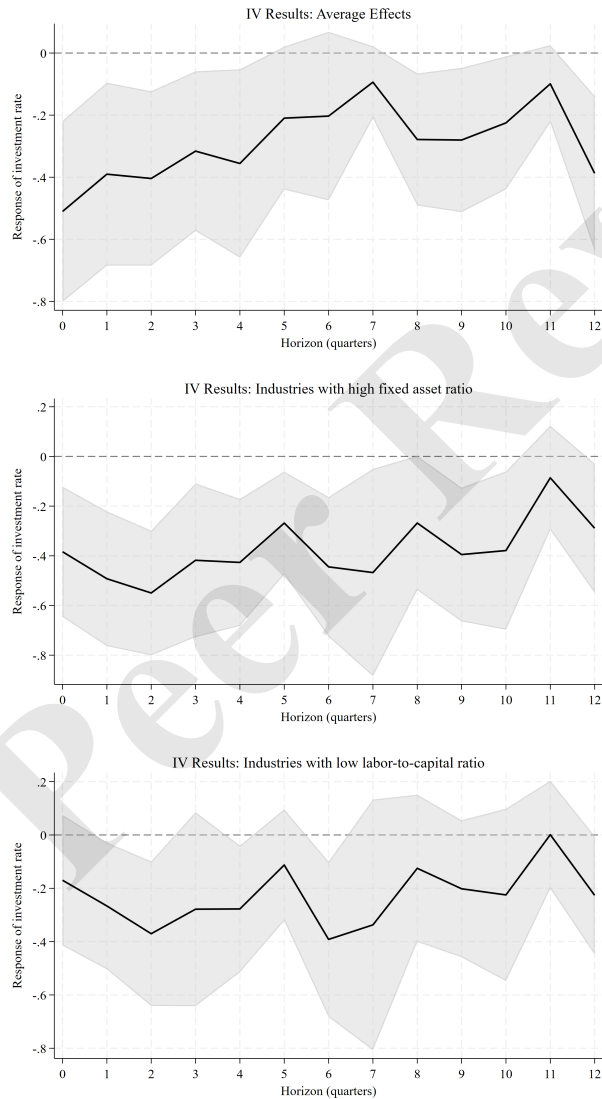
Date	Event	Reference
2020-04-14	NK fires several projectiles, presumed short-range cruise missiles, from Munchon	Ministry of Unification
2020-06-08	NK announces cutting all communication lines with South Korea	armscontrol association
2020-06-16	NK demolishes the inter-Korean liaison office in Kaesong	armscontrol association
2020-10-10	NK unveils new ICBM at military parade marking party's 75th anniversary	armscontrol association
2021-01-13	NK announces new weapons development plans (hypersonic, MIRV etc.) at 8th Party Congress	armscontrol association
2021-03-25	NK fires 2 unidentified projectiles into East Sea	Ministry of Unification
2021-03-25	NK tests KN-23 variant, first missile test under Biden administration	armscontrol association
2021-09-13	NK announces test of new long-range cruise missile	armscontrol association
2021-09-15	NK test-fires ballistic missiles from a train	armscontrol association
2021-09-28	NK tests hypersonic missile Hwasong-8	armscontrol association
2021-10-19	NK fires 1 SLBM from Sinpo into East Sea	Ministry of Unification
2022-01-05	NK conducts second hypersonic missile test	armscontrol association
2022-01-25	NK fires 2 cruise missiles	Ministry of Unification
2022-01-30	NK tests intermediate-range Hwasong-12, violating UN resolutions	armscontrol association
2022-02-27	NK fires 1 ballistic missile from Sunan into East Sea	Ministry of Unification
2022-03-05	NK fires 1 ballistic missile from Sunan into East Sea	Ministry of Unification
2022-03-16	NK launch from Sunan fails at 20km altitude	Ministry of Unification
2022-03-20	NK fires 4 presumed MLRS rounds into West Sea	Ministry of Unification
2022-03-24	NK fires 1 ICBM at lofted angle from Sunan into East Sea	Ministry of Unification
2022-05-04	NK fires 1 ballistic missile from Sunan into East Sea	Ministry of Unification
2022-05-07	NK fires SLBM from waters off Sinpo	Ministry of Unification
2022-05-12	NK fires 3 short-range ballistic missiles from Sunan into East Sea	Ministry of Unification
2022-05-25	NK fires 3 ballistic missiles from Sunan into East Sea	Ministry of Unification
2022-06-05	NK simultaneously fires 8 ballistic missiles from multiple sites	Ministry of Unification
2022-06-12	NK fires multiple rocket launcher	Ministry of Unification
2022-07-10	NK fires 2 MLRS rounds	Ministry of Unification
2022-08-17	NK fires 2 cruise missiles from Onchon into West Sea	Ministry of Unification
2022-09-25	NK fires 1 short-range ballistic missile from Taechon into East Sea	Ministry of Unification
2022-09-28	NK fires 2 ballistic missiles from Sunan into East Sea	Ministry of Unification
2022-09-28	NK fires 2 ballistic missiles from Sunchon into East Sea	Ministry of Unification
2022-10-01	NK fires 2 short-range ballistic missiles from Sunan into East Sea	Ministry of Unification
2022-10-04	NK fires 1 intermediate-range ballistic missile from Jagang Province eastward, overflying Japan	Ministry of Unification
2022-10-06	NK fires 2 short-range ballistic missiles from Samsok into East Sea	Ministry of Unification
2022-10-09	NK fires 2 short-range ballistic missiles from Munchon into East Sea	Ministry of Unification
2022-10-24	NK merchant ship crosses NLL northwest of Baengnyeong Island	Ministry of Unification
2022-10-28	NK fires 2 short-range ballistic missiles from Tongchon into East Sea	Ministry of Unification
2022-11-02	NK fires 10+ short-range ballistic and surface-to-air missiles from multiple sites into East and West Sea	Ministry of Unification
2022-11-02	NK fires 3 short-range ballistic missiles from Wonsan into East Sea, triggers air-raid alert on Ulleungdo	Ministry of Unification
2022-11-02	NK fires 4 short-range ballistic missiles from Jongju/Pyhyon into West Sea	Ministry of Unification
2022-11-03	NK fires 1 ICBM from Sunan into East Sea	Ministry of Unification
2022-11-03	NK fires 3 short-range ballistic missiles from Goksan into East Sea	Ministry of Unification
2022-11-05	NK fires 4 short-range ballistic missiles from Tongrim into West Sea	Ministry of Unification
2022-11-09	NK fires 1 short-range ballistic missile from Sukchon into East Sea	Ministry of Unification
2022-11-17	NK fires 1 short-range ballistic missile from Wonsan into East Sea	Ministry of Unification
2022-11-18	NK fires 1 ICBM from Sunan into East Sea	Ministry of Unification
2022-12-18	NK fires 2 medium-range ballistic missiles (MRBM) from Dongchang-ri into East Sea	Ministry of Unification
2022-12-23	NK fires 2 short-range ballistic missiles into East Sea	Ministry of Unification

Continued on next page

TABLE A.2 North Korea-related Events Timeline (2003-2024) – continued from previous page

Date	Event	Reference
2022-12-28	NK fires 3 short-range ballistic missiles from Junghwa into East Sea	Ministry of Unification
2023-01-01	NK fires 1 short-range ballistic missile from Yongsong into East Sea	Ministry of Unification
2023-02-18	NK fires Hwasong-15 ICBM from Sunan toward East Sea	Ministry of Unification
2023-02-20	NK fires 2 short-range ballistic missiles from Sukchon toward East Sea	Ministry of Unification
2023-02-23	NK conducts strategic cruise missile launch drill	Ministry of Unification
2023-03-14	NK fires SRBM from Jangyon into East Sea	Ministry of Unification
2023-03-16	NK fires ICBM from Sunan into East Sea	Ministry of Unification
2023-03-18	NK conducts simulated nuclear counterattack drill, fires SRBM from Dongchang-ri into East Sea	Ministry of Unification
2023-03-21	NK conducts weapons tests, including underwater attack drone and strategic cruise missile launch	Ministry of Unification
2023-03-27	NK fires SRBM from Junghwa into East Sea	Ministry of Unification
2023-04-13	NK fires IRBM-or-longer ballistic missile near Pyongyang into East Sea	Ministry of Unification
2023-05-28	NK launches satellite southward from Dongchang-ri	Ministry of Unification
2023-06-15	NK fires SRBM from Sunan into East Sea	Ministry of Unification
2023-07-12	NK fires Hwasong-18 ICBM from Pyongyang into East Sea	Ministry of Unification
2023-07-19	NK fires SRBM from Sunan into East Sea	Ministry of Unification
2023-07-22	NK fires several cruise missiles into West Sea	Ministry of Unification
2023-07-24	NK fires SRBM into East Sea	Ministry of Unification
2023-08-28	NK fires 2 SRBMs from Sunan into East Sea	Ministry of Unification
2023-09-02	NK CMC conducts simulated tactical nuclear attack drill, fires 2 long-range strategic cruise missiles	Ministry of Unification
2023-09-13	NK fires SRBM from Sunan into East Sea	Ministry of Unification
2023-11-21	NK launches spy satellite Malligyong-1, Kim Jong Un observes	Ministry of Unification
2023-12-18	Kim Jong Un observes launch of Hwasongpho-18 ICBM	Ministry of Unification
2024-01-14	NK claims successful test of hypersonic solid-fuel IRBM	Ministry of Unification
2024-01-24	NK conducts first test of new strategic cruise missile Pulhwasal-3-31 into West Sea	Ministry of Unification
2024-01-28	NK test-fires submarine-launched strategic cruise missile Pulhwasal-3-31, Kim Jong Un observes	Ministry of Unification
2024-01-28	NK conducts strategic cruise missile (Hwasal-2) launch drill into West Sea	Ministry of Unification
2024-02-02	NK fires several cruise missiles into West Sea	Ministry of Unification
2024-02-14	NK fires several unidentified cruise missiles from waters off Wonsan	Ministry of Unification
2024-03-18	NK fires 3+ short-range ballistic missiles into East Sea	Ministry of Unification
2024-04-02	NK fires 1 medium-range ballistic missile into East Sea, claims new hypersonic solid-fuel missile test success	Ministry of Unification
2024-04-22	Kim Jong Un directs simulated nuclear counterattack drill, ballistic missile launched	Ministry of Unification
2024-06-26	NK fires 1 unidentified ballistic missile into East Sea	Ministry of Unification
2024-07-01	NK fires 2 short-range ballistic missiles into East Sea	Ministry of Unification
2024-07-02	NK Missile Administration claims successful test of new tactical ballistic missile	Ministry of Unification
2024-09-12	NK fires several short-range ballistic missiles	Ministry of Unification
2024-09-18	NK fires several short-range ballistic missiles	Ministry of Unification
2024-11-05	NK fires short-range ballistic missile	Ministry of Unification

FIGURE A.4 Impulse response functions from local projection: 2SLS Results



Notes: IRFs to a 100% GPRNK shock over 12 quarters. Shaded bands denote 95% CIs.

A.3. Variable Description

TABLE A.3 Variable Description

Variable	Description	Frequency	Source
Investment expenditures	Capital expenditures plus R&D expenditures scaled by the lagged total assets	Quarterly	Value Search
Tobin's q	Market capitalization plus total debt divided by the book value of physical asset	Quarterly	DataGuide, Value Search
Leverage Ratio	The ratio of total debt to market capitalization	Quarterly	DataGuide, Value Search
Return on Assets (ROA)	The ratio of operating profit to total asset value	Quarterly	Value Search
Asset Size	Natural logarithm of total asset value	Quarterly	Value Search
Sales Growth	Growth rate of sales calculated by log differences	Quarterly	Value Search
Capital intensity	Value of fixed asset divided by value of total asset	Quarterly	Value Search
Labor to capital ratio	Natural logarithm of the ratio of the number of employees to physical asset value	Quarterly	Value Search
Labor-capital substitutability	Industry-level standard deviation of the labor-capital ratio	Quarterly	Value Search
Economic cooperation	Dummy variable indicating involvement in economic cooperation	-	Author's Compilation
Defense industry	Dummy variable indicating business related to defense industry	-	Author's Compilation
Composite index of leading indicators		Quarterly	Ministry of Data and Statistics
World merchandise trade (s.a.)		Quarterly	Netherlands Bureau for Economic Policy Analysis (CPB)

A.4. Smooth transition autoregression model for the GPRNK index

To examine whether the dynamics of the GPRNK index have changed gradually over time, we estimate a smooth transition autoregressive model in which the parameters evolve as a logistic function of deterministic time (Lin and Teräsvirta, 1994; Cho, 2018). This specification allows the persistence and long-run level of the series to shift smoothly between two limiting regimes without imposing an abrupt break date. Standard unit root tests reject a unit root in $y_t = \ln \text{GPRNK}_t$, and the Bayesian information criterion (BIC) selects an autoregressive order of two. We use the full monthly GPRNK series from 1995 to 2024, while the subsequent investment regressions are restricted by the availability of firm-level data. We estimate

$$y_t = (\phi_{10} + \phi_{11}y_{t-1} + \phi_{12}y_{t-2})[1 - G(z_t; \gamma, c)] + (\phi_{20} + \phi_{21}y_{t-1} + \phi_{22}y_{t-2})G(z_t; \gamma, c) + u_t, \quad (4)$$

where

$$G(z_t; \gamma, c) = [1 + \exp\{-\gamma(z_t - c)\}]^{-1}, \quad \gamma > 0, \quad z_t = t/T \quad (5)$$

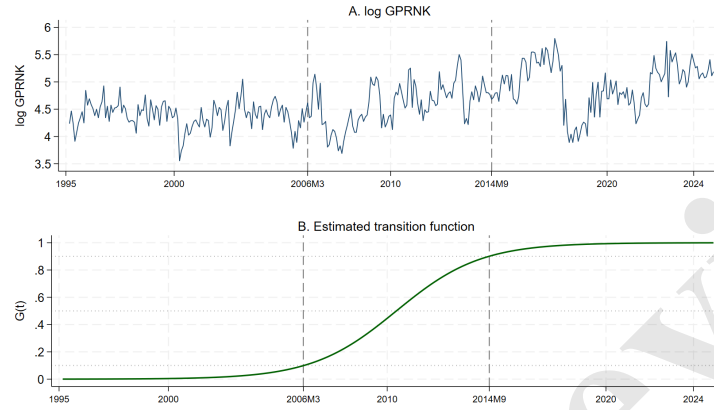
The transition variable z_t is normalized calendar time, so the model captures a single gradual shift between two autoregressive regimes. The parameter c locates the transition midpoint, while γ governs its speed. As $\gamma \rightarrow \infty$, the model approaches an abrupt break at c . This provides a parsimonious smooth time-varying autoregressive specification of structural change.

Because γ is unidentified under the null of parameter constancy, linearity is tested using the Taylor-expansion LM test of (Lin and Teräsvirta, 1994). Parameters are estimated by nonlinear least squares, concentrating out the autoregressive coefficients and numerically optimizing over (γ, c) .

The null hypothesis of parameter constancy is rejected ($LM = 26.11$, $p = 0.002$), providing evidence that the dynamics of the GPRNK index evolve over time. The estimated transition is gradual, with $\hat{c} = 0.515$, corresponding to a midpoint of 2010M6. Persistence, measured by the sum of the two autoregressive coefficients, increases from 0.59 in the early limiting regime to 0.83 in the late limiting regime. The implied long-run level of the index also rises substantially, approximately doubling between the two regimes.

To characterize the timing of the change, we define the transition interval as the period over which the fitted transition function rises from $\hat{G}_t = 0.1$ to $\hat{G}_t = 0.9$, yielding an estimated transition period from 2006M3 to 2014M9 (Figure A.5).

FIGURE A.5 Estimated Smooth Transition AR(2) Model for the GPRNK index



For the main subsample analysis, we use the end of this transition to divide the sample into observations through 2014Q3 and the post-transition period beginning in 2014Q4. To test whether the investment response differs across these periods, we estimate a pooled specification interacting GPRNK with the post-transition indicator.

$$Inv_{i,t+1} = \theta_1 GPRNK_t + \theta_2 (GPRNK_t \times Post_t) + \lambda Post_t + \mathbf{C}'_{i,t} \gamma + \alpha_i + \delta_q + \varepsilon_{i,t+1}, \quad (6)$$

where $Post_t$ equals one from 2014Q4 onward and zero otherwise. The coefficient θ_1 captures the effect of GPRNK prior to the transition, while θ_2 measures the change in this effect in the post-transition period. The post-transition effect is therefore $\theta_1 + \theta_2$. For the three periods, we estimate the following.

$$\begin{aligned} Inv_{i,t+1} = & \theta_1 GPRNK_t + \theta_2 (GPRNK_{t-1} \times Transition_t) \\ & + \theta_3 (GPRNK_t \times Post_t) + \lambda_1 Transition_t + \lambda_2 Post_t \\ & + \mathbf{C}'_{i,t} \gamma + \alpha_i + \delta_q + \varepsilon_{i,t+1}, \end{aligned} \quad (7)$$

where $Transition_t$ equals one for 2006Q1–2014Q3 and zero otherwise, and $Post_t$ equals one from 2014Q4 onward and zero otherwise.

The estimation results are reported in Table A.4. The negative effect weakens after the transition, with the post-transition estimate close to zero and statistically insignificant. The pooled tests reject the equality of the GPRNK effects across periods, confirming a significant attenuation in the investment response over time.

TABLE A.4 Pooled Interaction Tests across Transition Periods

	(1)	(2)
<i>Panel A. Pooled interaction regressions</i>		
<i>GPRNK</i>	−0.0050*** (0.0018)	−0.0109 (0.0066)
<i>GPRNK</i> × <i>Transition</i>		0.0065 (0.0069)
<i>GPRNK</i> × <i>Post</i>	0.0058*** (0.0020)	0.0116* (0.0067)
<i>Panel B. Implied GPRNK effects</i>		
Pre-transition	−0.0050*** (0.0018)	−0.0109 (0.0066)
Transition		−0.0043** (0.0019)
Post-transition	0.0007 (0.0007)	0.0007 (0.0007)
<i>Panel C. Tests of equality</i>		
Joint equality across periods (<i>p</i> -value)	0.004	0.016
Firm controls	Yes	Yes
Firm fixed effects	Yes	Yes
Quarter-of-year fixed effects	Yes	Yes
Observations	117,934	117,934
Adjusted <i>R</i> ²	0.0766	0.0767

Notes: Column (1) divides the sample into periods through 2014Q3 and from 2014Q4 onward. Column (2) distinguishes the pre-transition period (through 2005Q4), transition period (2006Q1–2014Q3), and post-transition period (from 2014Q4 onward), based on the smooth transition model estimation. Panel B reports the implied period-specific effects of GPRNK. Standard errors in parentheses are two-way clustered by firm and year-quarter. ****p* < 0.01, ***p* < 0.05, **p* < 0.10.

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Response to Referees' Comments

We are very grateful to the Referees for sharing very insightful comments and suggestions. We have taken your comments on board and revised the manuscript accordingly. Please find below our responses to each of your comments.

Major comments

1. Literature review

Reviewer 1-1: Discuss further about the literature. Using risk or uncertainty spillover effects to explain the mechanism.

⇒ We thank referees for this constructive feedback. We have expanded the discussion of related literature in the Introduction to clarify the transmission mechanism (paragraph 4, p.2). In particular, we now relate our findings to evidence on the transmission of political uncertainty to firm decisions (Gulen and Ion, 2016; Hassan et al., 2019) and on the spillover effects of geopolitical risk associated with proximity to military conflict (Federle et al., 2026). We also discuss recent evidence highlighting flexible inputs as an adjustment margin under uncertainty (Bloom et al., 2024), which provides further context for our heterogeneity results. These additions clarify how North Korea-related geopolitical risk can transmit to South Korean firms and why the resulting investment response varies with investment irreversibility and input flexibility.

2. Endogeneity concerns

Reviewer 1-2: The authors should use an appropriate IV model to address the endogeneity issue.

Reviewer 2-1: The identification strategy requires further justification. While geopolitical shocks originating from North Korea are plausibly exogenous, the paper does not fully explain why the GPRNK index can be interpreted as an exogenous measure of uncertainty rather than media attention or broader macroeconomic conditions. A stronger discussion of causality would improve the paper.

⇒ We greatly appreciate the helpful feedback. To address concerns about potential endogeneity, we have added an instrumental variable analysis and expanded the discussion of identification in the revised manuscript (paragraph 1, p. 5). Following Ha et al. (2022), we instrument GPRNK with a quarterly count of negative North Korea-related events, constructed from a hand-collected chronology of major military, nuclear, and related policy developments. The event list draws on official sources and is cross validated against existing event lists. Further details are provided in Appendix A.2.

The IV estimates remain negative and statistically significant and are slightly larger in magnitude than the baseline estimates (Table 3). The IV results for the heterogeneity analysis (Table 4) and local projections (Appendix Figure A.4) are also consistent with the baseline findings.

3. Economic interpretation of results

Reviewer 1-3: Interpret the magnitude of the impact. I suggest the authors to standardize the geopolitical risk index for ease of interpretation.

Reviewer 2-3: The economic significance of the results should be discussed more carefully. Although the estimated effects are statistically significant, the practical magnitude of the investment decline is not sufficiently contextualized relative to average investment rates or previous studies.

⇒ We are grateful for these comments. As described in Section 2, GPRNK is normalized so that its average over 1995-2016 equals 100. We have also expanded the discussion of economic significance in Section 4 (paragraph 3, p. 5). The baseline estimate implies that a doubling of GPRNK reduces quarterly investment by approximately 0.19 percentage points of assets, equivalent to about 14% of the average investment rate. We further contextualize this magnitude by referring to Wang et al. (2024), who report a similar decline relative to average investment for U.S. firms.

Reviewer 2-4: The interpretation of the post-2012 attenuation effect appears speculative. The argument that firms adapted to persistent geopolitical risk is interesting but not formally tested. Alternative explanations should be discussed or the interpretation should be stated more cautiously.

⇒ We thank the reviewer for this comment. We agree that our earlier interpretation of the attenuation was too strong because this mechanism is not directly tested. We have therefore revised the analysis and discussion in two ways. First, rather than imposing a pre-specified time split, we estimate a smooth transition model (Lin and Teräsvirta, 1994) for GPRNK that identifies a gradual transition ending in 2014Q3 (Appendix A.4). Second, we formally test for attenuation using pooled regressions that interact GPRNK with the transition-period indicators. The pooled tests reject equality of the GPRNK effects across periods, and the estimated effect becomes close to zero and statistically insignificant in the post-transition period (Table A.4).

We have also revised the main text of our manuscript to state the attenuation result more cautiously (paragraph 3, p. 6). In particular, we removed the earlier interpretation of the underlying channel, which was not directly tested, and refrain from attributing the attenuation to a specific mechanism.

4. Contribution of the index

Reviewer 2-2: The incremental contribution of the GPRNK index remains unclear. The manuscript introduces a new North Korea-specific geopolitical risk index, but it is not demonstrated whether this index provides information beyond existing geopolitical risk measures (e.g., Caldara and Iacoviello's Korea GPR index). A direct comparison would strengthen the contribution.

⇒ We thank the reviewer for this helpful comment. The Appendix already provides direct comparisons of the GPRNK index with both Caldara and Iacoviello's (2022) Korea GPR index and a GDELT-based index. In response to the reviewer's comment, we highlight the comparison with the Korea GPR index in the main text (paragraph 1, p. 4).

The main distinction is that the Korea GPR index primarily captures broader and globally salient geopolitical risks, whereas GPRNK is designed to capture persistent North Korea specific tensions, including recurrent missile activity and tactical nuclear signaling. Consistent with this distinction, the two indices are only moderately correlated (0.33). We have revised the manuscript to make this incremental contribution more explicit and refer readers to the Appendix for the detailed comparison.

5. Policy implications

Reviewer 1-4: Briefly suggest several policy implications.

⇒ Thank you for this helpful suggestion. We have added a brief discussion of policy implications in the conclusion (paragraph 5, p. 6). This addition helps place our findings in a broader context of heightened geopolitical tensions and fragmentation. In particular, we discuss the implications of the stronger response of capital-intensive firms and the mitigating role of greater input flexibility, highlighting the importance of a stable and predictable policy environment for long-term investment and of reducing frictions in input adjustment to strengthen firms' resilience to geopolitical shocks.

Minor comments

1. The construction of the GPRNK index relies on keyword searches, and the sensitivity of results to alternative keyword selections is not discussed.

⇒ We are grateful for this comment. We have added a sensitivity analysis using an alternative index, GPRNK-raw, constructed with the same keyword searches but without excluding articles containing category-specific negation terms. GPRNK-raw is highly correlated with the baseline GPRNK index (0.98), indicating that the index is not sensitive to this aspect of keyword selection. The comparison is now reported in Section 2 and Appendix Figure A.3.

2. The proxy for labor-capital substitutability may not fully capture technological substitution possibilities at the firm level and deserve further discussion.
 - ⇒ We thank the reviewer for this important point. We agree that the industry-level dispersion of the labor-capital ratio should not be interpreted as a direct measure of firm-level technological substitutability. As in Leahy and Whited (1996) and Wang et al. (2024), the measure is intended to capture greater flexibility in adjusting labor relative to relatively fixed capital. However, as an industry-level measure, it may not fully reflect firm-specific substitution possibilities. We have clarified this limitation in the revised manuscript (footnote 2, p. 4).
3. The manuscript would benefit from a brief explanation in the main text of how the GPRNK index is calculated, rather than leaving most methodological details to the appendix.
 - ⇒ Thank you for this suggestion. We have revised Section 2 to provide a brief description of the GPRNK construction in the main text. The revised manuscript explains the news sources and outlet coverage, the classification of North Korea-related articles into negative and positive categories, the treatment of negation terms, and the aggregation and normalization of the index. We also report a sensitivity check using an alternative index constructed without excluding negation terms. Further details on the keyword sets and index construction remain in Appendix A.1.
4. There are a few editorial issues, including formatting inconsistencies and typographical errors (e.g., "Calender quarter dummies" should be "Calendar quarter dummies").
 - ⇒ Thank you for the comment. We have corrected the typographical and formatting errors throughout the manuscript.

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